Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

AI/ML engineer with **8+ years of production Python** and hands-on experience across the full GenAI stack - LLMs, prompt engineering, RAG pipelines, and agentic workflows. Built **AgentProbe** end-to-end: multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama), RAG pipelines (ChromaDB), 3-agent LangGraph system (Orchestrator, RAG Tracer, Auto Fix), and a 5-metric evaluation framework - using **Claude Code and GitHub** daily throughout development. AWS production experience (S3, EC2, RDS, Glue) at scale across 1.6M financial records and 100K+ users. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

Python (8+ years production) | LLM Integration (Anthropic, OpenAI, Groq, Ollama) | Prompt & Context Engineering | RAG Pipelines & Vector Search (ChromaDB) | Agentic AI (LangGraph, Multi-Agent) | Token Management & Session State | LLM Evaluation Framework | AWS (S3, EC2, RDS, Glue) | Claude Code / GitHub Copilot / CI/CD | FastAPI / Docker / Microservices | Production at Scale (1.6M records, 100K+ MAU) | TypeScript / JavaScript / SQL

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Applied **Python, PyTorch, and OpenAI API** to automate data transformation across **1.6M financial records** - production AI on regulated data with careful output validation and quality monitoring.
- Built and delivered **20+ full-stack backend modules** (Python, PostgreSQL, JavaScript) across **120+ Agile sprints** for **100K+ monthly active users** - 99.5% uptime over 6+ consecutive months.
- Led Docker-based containerization for **30%+ of microservices**, cutting execution time by **50%**; improved production query performance by **30%** across 200+ SQL queries.
- Migrated client data to **AWS S3** with Libsodium-encrypted pipelines - **100% PHI compliance, zero security incidents** across 100K+ users; managed services across EC2, RDS, and Glue.
- Built **5+ internal platform tools** adopted across squads, reducing resolution time by **40%**; enforced secure coding standards and peer code review practices across all releases.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Mentored 30+ students in Python and database systems; designed supplementary exercises and drove iterative quality improvement across the course.

PROJECTS

AgentProbe: LLM + Agentic AI Platform - RAG, Multi-Agent, Evaluation OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD

End-to-end AI/ML engineering platform built with Claude Code and GitHub, demonstrating the exact stack this role requires. **LLM integration**: multi-provider routing across Anthropic, OpenAI, Groq, Ollama - dynamic selection by task type, latency, and capability. **Prompt & context engineering**: context window management, short-term vs. long-term memory patterns, token budget optimization across multi-step agent runs. **RAG & vectorization**: ChromaDB embeddings pipeline with retrieval quality scoring and hallucination detection. **Agentic AI**: 3-agent LangGraph system (Orchestrator, RAG Tracer, Auto Fix) with autonomous multi-step reasoning, tool calls, and inter-agent state handoffs. **Evaluation framework**: 5-metric LLM-as-judge engine (factual accuracy, hallucination, retrieval, completeness, compliance) with CI/CD regression gates - A/B testing for AI quality. **Platform**: 15-endpoint FastAPI backend + React dashboard. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: Clinical NLP Platform - LLMs + BERT + FHIR APIs

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings · NumPy · Pandas

Production NLP system routing natural language queries to domain-specific FHIR APIs via LLMs, BERT embeddings, and spaCy - HIPAA-compliant RBAC. **~60% retrieval time reduction.** Demonstrates building GenAI solutions for professional workflows with strict compliance requirements. *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: Predictive ML Pipeline - 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Feature Engineering · GitHub Pages

Full ML lifecycle: feature engineering → model selection → evaluation → deployment. **85%+ accuracy.**

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML systems, experimental design, production data pipelines

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

AI & GenAI: LangChain, LangGraph, Multi-Agent Orchestration, LLM APIs (Anthropic, OpenAI, Groq, Ollama), Prompt Engineering, Context Engineering, Token Management, RAG Pipelines, ChromaDB, Vector Search, LLM-as-Judge Evaluation, CI/CD Eval Gates

ML & Data: PyTorch, scikit-learn, BERT, spaCy, Hugging Face, NumPy, Pandas, Feature Engineering, Statistical Modeling

Cloud & DevOps: AWS (S3, EC2, RDS, Glue), Azure, Docker, FastAPI, Flask, React, PostgreSQL, MySQL, RabbitMQ, CI/CD (GitHub Actions)

Tooling: Claude Code, GitHub Copilot, Git, JIRA, Agile/Scrum, TDD/BDD